

Q.5 The type of drag, which associated with the shape of aircraft is:

- a) Skin friction drag b) Form drag
- c) Interference drag d) Induced drag

Q.6 The angle between RAF and chord line called:

- a) Incidence angle b) AOA
- c) Stall angle d) Wash-Out angle

Q.7 The axis that extends from Nose section to Tail section of an aircraft is called:

- a) Lateral axis b) Chord wise axis
- c) Longitudinal axis d) Vertical axis

Q.8 One of the following is not considered as a factor of generating lift:

- a) Airfoil shape b) Air velocity
- c) Air temperature d) Air density

Q.9 The type of airfoil that used in Rotary aircraft is:

- a) Symmetrical airfoil b) Asymmetrical airfoil
- c) Camber airfoil d) NASA airfoil

Q.10 The ratio of air flow velocity to the speed of sound is called:

- a) Reynolds number b) Avogadro's number
- c) Mach number d) Pascal law

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What happens to the density of air as we go up?

Q.12 What is the function of flap?

Q.13 What is the cause of lift in an airplane?

Q.14 What is the purpose of sweeping wings forward?

Q.15 How does an airplane fly?

Q.16 How does the temperature vary in troposphere?

Q.17 What are different types of flows?

Q.18 Name the different secondary controls.

Q.19 How does the helicopter move forward?

Q.20 What is the main function of trim tab control?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the use of major parts of an airplane?

Q.22 What are the characteristics of transonic flows?

Q.23 What are the different methods of control boundary layer.

Q.24 What is the function of tail in stability of the airplane?

Q.25 What is drag and what type of drags are encountered by aircrafts?

Q.26 What is a mach number? Name the different regimes of flow speed based on Mach Number?